



Employee Attendance Automation System (Google Sheets)

Project Overview

Built a dynamic Employee Attendance Automation System in Google Sheets to simplify attendance recording and HR reporting. The system automatically calculates attendance metrics and stores records in a structured database, reducing manual work and improving reporting accuracy.

Project Objective

- Automate employee attendance entry.
- Reduce manual calculations.
- Store attendance records in a structured database.
- Generate real-time HR analytics through a dashboard.
- Practice automation using Google Sheets and Macros.

Tools Used

- Google Sheets
- Google Sheets Macro
- Google Sheets Formulas

- Dashboard Design

Project Structure

1. Employee List

Stores master employee information

Employee ID	Employee Name	Department	Shift Start	Shift End
EMP001	Rahim Ahmed	Sales	9:00 AM	6:00 PM
EMP002	Karim Hossain	HR	9:30 AM	6:30 PM
EMP003	Hasan Ali	Finance	8:30 AM	5:30 PM
EMP004	Nusrat Jahan	Marketing	10:00 AM	7:00 PM
EMP005	Tanvir Islam	IT	9:00 AM	6:00 PM
EMP006	Farzana Akter	Accounts	8:00 AM	5:00 PM
EMP007	Mahmud Hasan	Operations	9:30 AM	6:30 PM
EMP008	Sadia Rahman	Customer Support	10:00 AM	7:00 PM
EMP009	Imran Kabir	Procurement	9:00 AM	6:00 PM
EMP010	Mehedi Hasan	Logistics	9:00 AM	6:00 PM
EMP011	Ayesha Siddika	HR	9:30 AM	6:30 PM
EMP012	Rifat Chowdhury	Sales	9:00 AM	6:00 PM
EMP013	Shakil Ahmed	IT	9:00 AM	6:00 PM
EMP014	Nafisa Islam	Finance	8:30 AM	5:30 PM
EMP015	Arif Hossain	Operations	9:00 AM	6:00 PM
EMP016	Sumaiya Akter	Marketing	10:00 AM	7:00 PM
EMP017	Fahim Ahmed	Accounts	8:00 AM	5:00 PM
EMP018	Tania Sultana	Customer Support	10:00 AM	7:00 PM
EMP019	Sabbir Rahman	Procurement	9:00 AM	6:00 PM
EMP020	Jannatul Ferdous	Logistics	9:00 AM	6:00 PM

Fields

- Employee ID
- Employee Name
- Department
- Shift Start
- Shift End

Purpose

Acts as the master data source for the attendance system.

2. Attendance Entry Form

Created a user-friendly attendance form where users only enter:

- Date
- Employee ID
- In Time
- Out Time
- Remarks

Other fields are generated automatically.

Employee Attendance Entry Form	
Date	
Employee ID	
Employee Name	
Department	
In Time	
Out Time	
Work Hours	0.00
Late Minutes	
Overtime Hours	
Status	
Remarks	
SUBMIT	

3. Automated Features

Employee Name

Automatically retrieved using XLOOKUP.

Department

Automatically retrieved using XLOOKUP.

Work Hours

Automatically calculated from:

$(\text{Out Time} - \text{In Time}) * 24$

Converted into decimal hours.

Late Minutes

Calculated by comparing:

Actual In Time
vs
Shift Start Time

Returns zero if the employee arrives on time.

Overtime Hours

Calculated by comparing:

Actual Out Time
vs
Shift End Time

Only overtime beyond the assigned shift is counted.

Status

Automatically assigned as:

- Present
- Late
- Half Day

Based on attendance rules.

4. Attendance Database

Every submitted attendance record is stored automatically in the database.

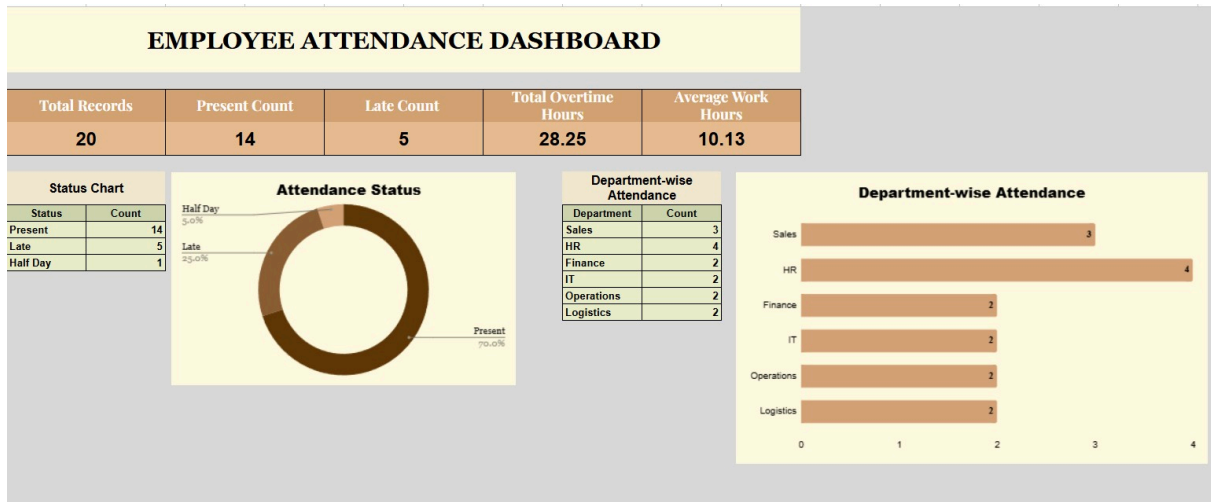
A	B	C	D	E	F	G	H	I	J	K
Date	Employee ID	Employee Name	Department	In Time	Out Time	Work Hours	Late Minutes	Overtime Hours	Status	Remarks

Database Columns

- Date
- Employee ID
- Employee Name
- Department
- In Time
- Out Time
- Work Hours
- Late Minutes
- Overtime Hours
- Status
- Remarks

5. Dashboard

Built an interactive HR dashboard using attendance data.



KPI Cards

- Total Records
- Present Count
- Late Count
- Total Overtime Hours
- Average Work Hours

Charts

- Attendance Status Distribution
- Department-wise Attendance

Key Formulas Used

Employee Name

XLOOKUP ()

Department

XLOOKUP ()

Work Hours

$(\text{Out Time} - \text{In Time}) \times 24$

Late Minutes

$(\text{In Time} - \text{Shift Start}) \times 1440$

Overtime Hours

$(\text{Out Time} - \text{Shift End}) \times 24$

Dashboard KPIs

COUNTA()
COUNTIF()
SUM()
AVERAGE()
ROUND()

Business Benefits

- Eliminates repetitive manual calculations.
- Standardizes attendance recording.
- Provides instant attendance summaries.
- Supports HR decision-making with visual dashboards.
- Reduces reporting time.